Leakage Prevention and Real-Time Internal Detection in Pipelines Using a Built-In Wireless Information and Communication Network

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Summary

A series of recent pipeline leakage incidents created severe societal concerns to a point of impeding, or even completely preventing, building new pipelines in North America. Various systems have been proposed to identify and locate leakages. However, despite the fact that pipelines remain the safest means of oil and gas transportation, incidents still persist and pipeline acceptance from the public has become compromised. In order to address the need for early leakage detection, while providing comprehensive leakage prevention, a novel pipeline system is proposed. This concept builds on the already existing pipe-in-pipe design by segmenting the pipeline system with segmentation rings and embedding a linear wireless network in the annular airgap between the two pipe layers. Presence of fluid in the case of a leakage into the interpipe space causes degradation of the wireless network to a point of interrupting the communication in a particular pipeline segment well before any external leak occurs. Thus, the internal leak is detected in real time by an external central unit connected to the linear wireless network, as demonstrated with a 6 ft 8 in. experimental pipeline setup.

Introduction

Pipelines as Major Means of Oil and Gas Transportation. Oil and gas are transported across the United States by means such as pipelines, water carriers (tankers and barges), motor carriers (trucks), and railroads (trains) (National Transportation Statistics 2017). Among these, pipelines are considered the most efficient and cost-effective and the safest way for transporting high-risk fluids (US Department of Transportation 2010; Furchtgott-Roth 2012; About Pipelines 2016). The percentage of crude oil and petroleum products transported by pipelines in the US increased significantly during the last three decades, rising from 47.2% in 1985 to 70.1% in 2009 (National Transportation Statistics 2017).

Pipelines are divided into three major categories, on the basis of the nature of the transported fluid: crude oil, oil-based products, and natural gas (National Transportation Statistics 2017). Alternatively, they can be classified on the basis of their location: underground, above ground, underwater, and possibly a mix of any of these categories. The cost of construction varies among these categories and among different geographical areas (Rui et al. 2011). In addition, the cost per mile for building a pipeline generally decreases as the length of the pipeline increases (Smith 2016).

The US Federal Energy Regulatory Commission (FERC) annual report from 2015 shows that US interstate pipeline mileage for gas has not significantly increased in the past decade (US Federal Energy Regulatory Commission 2015). Annual Report Mileage released in 2017 by the Pipeline and Hazardous Materials Safety Administration (PHMSA) of the US Department of Transportation stated that the overall percentage of annual increase from 2005 to 2014 was greater for oil pipelines than for gas pipelines (US Department of Transportation 2017).

Oil pipelines are classified as carrying crude oil, refined products, or highly volatile liquids (HVL). Crude oil pipelines transport heavy oil containing diluted bitumen (dilbit) from exploration sites to refineries. Refined product pipelines connect refineries to distribution terminals, usually thousands of miles apart; from there, refined products are generally transported to their final destinations using alternative means of transportation, such as barges and trucks. HVL pipelines are considered a special subset due to the nature of the fluid they transport, yet may include the previous two types of pipelines—crude oil, and refined products.

Gas pipelines are classified as gathering lines, transmission lines, and distribution lines. Gathering lines are located close to onshore production sites; they gather raw natural gas from production wells and transport it toward larger transmission pipelines. Transmission pipelines stretch thousands of miles across the country to transport natural gas toward distribution lines. Finally, distribution lines distribute natural gas to homes and businesses in thousands of communities across the country. The wall thickness of gas pipelines depends on the operating conditions, in particular, on operating pressures. Distribution lines commonly operate at pressures much lower than gathering lines, thus they tend to be of thinner walls. In addition, depending on the liquidity or the dryness of their content, gas pipelines have to comply with different requirements.

Among oil pipelines, crude oil pipelines exhibit the largest increase between 2005 and 2015, peaking at an almost 10% yearly increase in 2014/2015. Among gas pipelines, gathering lines experienced a significant decrease in mileage throughout most of the same 10-years span. Meanwhile, transmission and distribution lines had small rates of increase in most years, varying from 0 to 3%. Nearly the entire mainline pipeline system in the US is buried, while other pipeline components such as pump stations are installed above ground to provide easy access (Pipeline 101 2017). The length of these lines varies from as little as a mile to more than 1,000 miles. Some pipelines, however, are built in environments with very special characteristics. For example, the TransAlaska Pipeline was built above ground with built-in bends and teflon shoes lying on top of slider beams (US Geological Survey 2003). This design allows the pipeline to move during an earthquake, which is common around the Denali Fault area where it was installed, and it prevents the pipeline from crashing. Several long underwater pipeline systems have been installed and are in use for different applications around the world (Mohamed et al. 2011).

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For example, the Langeled Pipeline is one of the longest, as it extends for 725 miles and transports natural gas from the Ormen Lange Field in Norway to the Easington Gas Terminal in England under the North Sea (Solberg and Gjertveit 2007). Additionally, underwater pipelines are extensively encountered in the Gulf of Mexico (GOM), which hosts approximately 30,000 miles (Joyce 2006).

Pipe-in-Pipe (Double Pipe) Systems. Pipe-in-pipe and bundled pipeline systems currently allow petroleum products to be transported with a high level of thermal insulation and, therefore, they are widely used in the oil and gas industry (Zheng et al. 2012). A pipe-in-pipe flowline construction is normally selected because of its thermal performance, but it comes at the cost of complex fabrication and installation. The layer in between the two solid pipes is usually filled with insulation material, e.g., nanogel, in order to limit the U-value of the flowline (Hausner and Dixon 2002; Zheng et al. 2012; Dixon 2013). In deepwater environments, pipe-in-pipe systems are proven to be a reliable technology for insulated transportation of wellbore fluids (Hausner and Dixon 2002). A pipe-in-pipe system consists of two layers of pipe concentrically positioned, with the annular space often used to place an insulation material of high efficiency (Dixon 2013). In terms of thermal insulation, this approach is more efficient than a simple coating of single pipes; however, it naturally comes at a higher cost.

Along with the original interest in pipe-in-pipe designs for thermal insulation, this technology clearly offers the advantage of having a secondary containment over single-layer pipelines (US Department of Interior 2000). Leak containment and protection of outer pipes are achieved with single or multiple inner pipes bundled together inside one outer pipe (Ngiam et al. 1998). Use of pipe-in-pipe and bundled pipelines started to grow during the last decades of the 20th century and approximately 87% of them were insulated. Trends show that during the last 20 years, a yearly average of 150 km (93 miles) of pipe-in-pipe systems are laid undersea globally, with the majority at the North Sea and in North America (Dixon 2013). In fact, an estimated 1,700 miles of pipe-in-pipe systems have been installed undersea since the beginning of the 21st century. Additionally, the average depth under water of these pipelines rose from 200 m in the late '90s to 1200 m in the late 2010s. While providing a higher level of protection for a pipeline, the outer layer of pipe-in-pipe designs increases the cost of construction and maintenance of the entire pipeline system. Additionally, internal inspection of the outer layer is not trivial, as the pipe-in-pipe structure is naturally more complex than a single-layer pipeline and does not allow the entrance of pigging devices in the annular space between the two pipe layers. However, the existence of the annular interpipe space creates the possibility of adding leak detection devices to monitor the entire dynamics of the pipeline (Booles 2005).

Leakage Detection and Prevention in Pipe-in-Pipe System. A possible way to implement a monitoring system for leak prevention and leak detection in a pipeline has been proposed before (Booles 2005). In this system, a liquid-leak-detection device is added to monitor the liquid dynamics and to detect any leakage before it spills into the external environment. However, a longer version of this pipe system is not feasible, as the monitoring devices are added externally to the pipe system instead of being installed within. Additionally, utilization of a wireless network to integrate the collected information with an external central unit capable of alerting a human operator of such an event is not envisioned. Most importantly, any leakage in the interstitial space between the embedded pipes is not immediately detected because there is a distance between the location of the leak and the leakage detection device, and the exact location of the leak cannot be detected.

Recent approaches in pipe system monitoring demonstrate that employing a wireless sensor network (WSN) is very beneficial (Sadeghioon et al. 2014; Sheltami et al. 2016), because in the last 20 years, sensors and wireless devices have become cheaper and more accessible (Pottie and Kaiser 2000; Chong and Kumar 2013). The most significant motivations for introducing this technique include: (a) reducing human inspection, (b) real-time monitoring, and (c) reducing both economic losses and environmental pollution (Al-Kadi et al. 2013). A WSN can be used for monitoring water flow in a pipe system and detecting any leaks (Jeffries et al. 2008). However, WSNs can be quite sophisticated to operate, especially under wide temperature and pressure changes, which may result in operational delays and late detection. Because this system reacts to leaks that are already occurring, it cannot fully address the economic and environmental issues caused by pipeline leaks. Technologies that are able to detect leaks are essential, but combining them with leakage prevention capabilities makes them far more attractive. Implementing a wireless information and communication network (WICN) along a pipeline system is feasible, and has been suggested before (Meribout 2011; Al-Kadi et al. 2013; Sheltami et al. 2016). However, combining this feature with a pipe-in-pipe system design might not be a straightforward task. Positioning electronic devices within the inner pipe layer could lead to the transported fluid damaging the embedded electronic devices after long exposure to the dynamic fluid flow, and can also degrade the quality and integrity of any wireless communications. An alternative system has been presented, where the pipe is composed of different layers and the wireless electronic devices are placed in a nonconductive coating layer of the pipe (Jensen and Grythe 2014). In this technology, the communication scheme is based on a waveguide technique and relies on having electrically conductive material in the layers adjacent to an electrically nonconductive coating. Such electrically conductive material can be built into the pipe, or it can be an external medium, such as salty seawater. This approach is both complex to implement and not reliable, especially for fluids in gaseous phase. Moreover, this design only provides a wireless communication arrangement, not leakage detection (or prevention). Therefore, any leak in the electrically nonconductive coating is not immediately detectable, because the integrity of the electrically nonconductive coating is not immediately disturbed. The radio frequency conductivity of the waveguide is affected only when the electrically nonconductive coating becomes almost totally eroded by the leak, introducing a delay in the leak detection procedure, particularly if the pipe system is long. Finally, the location of the initial leak remains unknown. Similarly, implanting the interpipe space of a pipe-in-pipe system with leakage detection and various other sensors does not eliminate the danger of external leakage. In fact, a recent incident in Alberta, Canada, involving a modern pipe-in-pipe structure emphasized this point quite dramatically (McMurray et al. 2018).

As a result of all these research and development (R&D) considerations, it is widely accepted that double-layer pipelines are key to efficient and reliable leakage detection (US Department of Interior 2000; Hutchinson et al. 2007; Wilson 2009; Jax 2012; Baylot et al. 2014). Several patents have been issued on this topic and several papers have been published, describing the common idea of having a first pipe layer surrounded by a second pipe layer for added protection. However, none of these technologies provide a concept for an entirely leakless pipeline, capable of containing small initial leaks into a limited segment of the interpipe space, while at the same time informing the external world of the occurrence. While leakage detection is important and much explored, preventing a small internal leakage from developing into a more dangerous and potentially damaging external spill is, in our opinion, of much greater importance. Focus, therefore, should be devoted to preventing leaking fluid from reaching the external environment by creating a pipeline system that never leaks, and present-day industrial pipeline standards should be modified to call for *complete prevention from external leaks,* rather than merely include leakage detection.

Aim of this Study. The aim of this study is to present and test a realistic concept of a completely leakless pipeline, which would not only prevent external spills during its normal (non force majeure) operation, but would also contain small initial leaks into a limited interpipe space, so that a quick repair of the initially leaking pipeline segment can rapidly and efficiently follow, before this small initial leak can grow into a more significant one that can jeopardize the integrity of a larger part of the pipeline and even threaten its breakdown, resulting in external spilling into the surrounding environment. Thus, in this work we propose and laboratory-test a segmented pipe-in-pipe system, which embeds adjacent embedded wireless information and communication stations (WICS) delimiting each segment from both sides. The interpipe space in this initial design proposal is left air-filled. If the interpipe space is to be filled with a liquid or gaseous insulation filler, such as specific aerogel in order to increase the overall U-value of the pipeline, in order for the proposed method to be effective, this filler has to have high wireless communication signal conductivity, ideally, similar to the one of air. However, in this study we do not explore this option. At this time the proposed method would not work if the interpipe space is filled with a solid-state insulation material, even if the latter would be conductive to wireless communication signals. The essential feature of the proposed method is to offer real time detection of any leak occurring from the inner pipe into the interpipe space between two adjacent WICS far before this leak exits the outer pipe. Thus, with respect to the external environment surrounding the pipeline system, the technique offers *complete prevention from external leaks*.

Methods

Pipe-in-Pipe Design. The dimensions of pipes are usually specified by four measures: inner diameter (ID), outer diameter (OD), wall thickness (WT), and length (L). Due to conventions in North America, all measures are given in United States customary units, i.e., feet and inches. **Fig. 1** illustrates the basic measurements of a pipe. The three measures shown in the coaxial view are not always given. Instead, two out of the three measures in the coaxial view (ID, OD, and WT) are sufficient to calculate the missing one, since they are related by the following equation:

$$\mathrm{OD} = \mathrm{ID} + 2\mathrm{WT}. \tag{1}$$

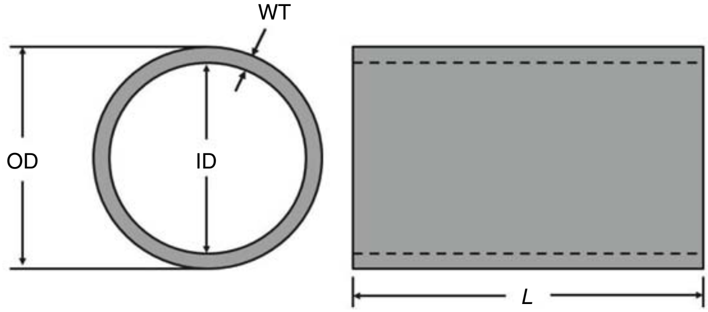

Fig. 1—Inner diameter (ID), outer diameter (OD), and wall thickness (WT) of a pipe.

Pipe-in-pipe design is a concept that includes one pipe layer being surrounded by another pipe layer. The internal diameter of the protective layer is larger than the outer diameter of the main pipe, allowing an airgap to be defined between the two pipe layers. The main pipe is responsible for transporting fluid, whereas the protective layer increases security. If a rupture in the main pipe causes leakage, the protective layer is able to contain the leakage, preventing the fluid from reaching the external environment. The two pipe layers are positioned concentrically so that the airgap maintains an annular shape. The coaxial view illustrates the ID, OD, and WT of each pipe layer, and indexes M and P identify the main pipe and the protective layer, respectively **(Fig. 2).**

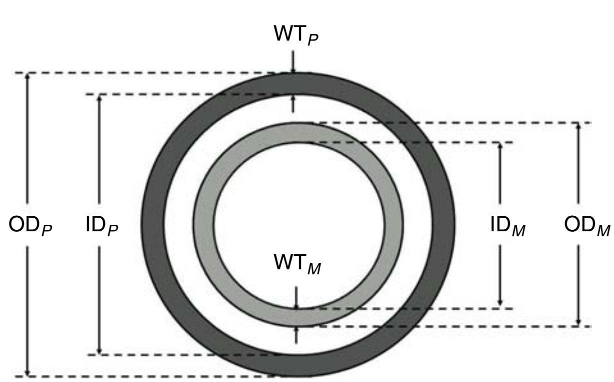

Fig. 2—Measurements of pipe layers in a pipe-in-pipe design.

Segmentation Rings. To concentrically position the main pipe with respect to the protective layer, segmentation rings are used within the annular airgap. The rings are custom-designed so that their width corresponds to the width of the annular airgap. These segmentation rings have the key purpose of segmenting the annular airgap of the double-walled pipeline, and represent the main innovation proposed in this paper. In the case of a leak from the main pipe into the annular airgap, the segment in which the rupture is located starts to flood. This segment temporarily contains the leaking fluid and helps the built-in wireless detection system to identify such leakage before it spreads to adjacent segments. One pair of segmentation rings defines a pipe apparatus, and interconnected pipe apparatuses would, then, build the entire pipeline.

Each segmentation ring has two predesigned openings, a larger one at the bottom and a small one at the top. The first predesigned opening is kept at the bottom of the annular airgap and hosts the electronic devices that establish a wireless information and communication stations, forming a linear network along the annular airgap. The angular cut of the first predesigned opening is determined on the basis of the dimensions of the devices that need to be placed within it. Ideally, the angular cut should not exceed 90° so that the stability of the structure is not compromised. The second predesigned opening is a small groove in the very top of each segmentation ring, which serves as a pressure relief valve if the annular airgap in any given segment starts to flood. Such pressure relief is important, as the interdependency between changing temperatures and pressures can make the environment in the airgap delimited by two consecutive segments quite dynamic.

As described in **Fig. 3,** each segmentation ring has an annular shape, with an outer diameter OD_R, an inner diameter ID_R, and an angular cut θ at the bottom. The length of the segmentation ring L_R depends on the size of the devices hosted in its predesigned opening.

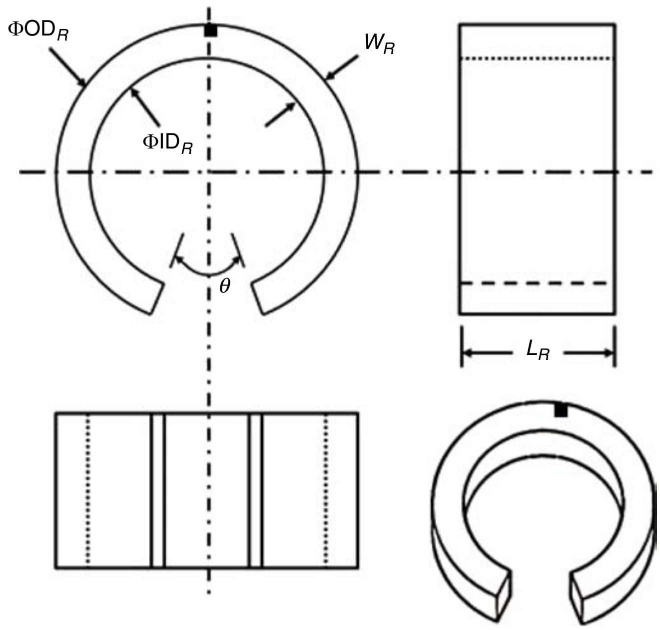

Fig. 3—Segmentation ring. The lower opening of angle θ is clearly seen, while the small groove of the upper opening is denoted with a black square.

In order to fit the segmentation rings within the annular airgap between the two pipe layers, the outer diameter of the segmentation ring is equal to the inner diameter of the protective layer, and the inner diameter of the segmentation ring is equal to the outer diameter of the main pipe:

$$OD_R = ID_P, \quad (2)$$

and

$$ID_R = OD_M. \quad (4)$$

Therefore, the equality between the width of the annular airgap and the width of the segmentation ring is guaranteed, as

$$W_A = W_R, \quad (4)$$

where W_A is the width of the annular airgap and W_R is the width of the segmentation ring, given by

$$W_R = (OD_R - ID_R)/2. \quad (5)$$

Fig. 4 illustrates one pipe apparatus containing two segmentation rings. Although only the segmentation ring at the right extremity of the pipe apparatus is visible, another one is placed at the left extremity of the pipe apparatus. The visible segmentation ring is displayed in light orange, the main pipe is displayed in light gray and the protective layer is displayed in dark gray. It would be beneficial for the segmentation rings containing the WICS to be located in the connectors between successive pipes in the entire pipeline system. Thus, in case of a leak from the inner pipe into the interpipe space, the entire leaking pipe can be easily replaced.

Fig. 4—Pipe apparatus containing main pipe, protective layer, and segmentation rings.

Double-layer pipelines have been proposed in the literature and have been in the public domain for more than 20 years (Hutchinson et al. 2007; Jeffries et al. 2008; Jax 2012); however, these designs do not envision using segmentation rings. Without this feature, when fluid leaks from the main pipe into the annular airgap, it may spread freely along the annular airgap until it reaches a crack in the protective layer, thus escaping into the external environment.

Embedded WICN and Internal Leakage Monitoring. The annular airgap is not only used for leakage prevention in the pipeline, but also for leakage detection purposes. A WICN is embedded within the annular airgap of the pipeline and is used to monitor and detect the presence of an undesired fluid (leakage) within the annular airgap. At least one external wireless station (EWS) belonging to the WICN and connected to at least one central unit is responsible for the monitoring task. Data received at this EWS is sent to dedicated software running in the central unit for further processing or emergency handling. This monitoring software operates in real time and identifies the location of the leak in the pipeline when such an event occurs. It can also trigger an alarm or even reduce pipeline pressure if need be in a fully automated closed-loop control system.

Since the WICN is located entirely inside the pipeline, an EWS must be added to the network in order to collect all the information packets from the last WICS in the chain. **Fig. 5** illustrates two possible implementations: (a) one EWS is connected at the end of a unidirectional linear WICN; and (b) two EWSs are connected, one at each end of a bidirectional WICN. Each EWS is connected to a central unit that processes all received data and keeps a log of the same. In the case where two EWSs are connected to two different central units, the communication between these two central units can be established through any other kind of network, such as the Internet, allowing information to be shared.

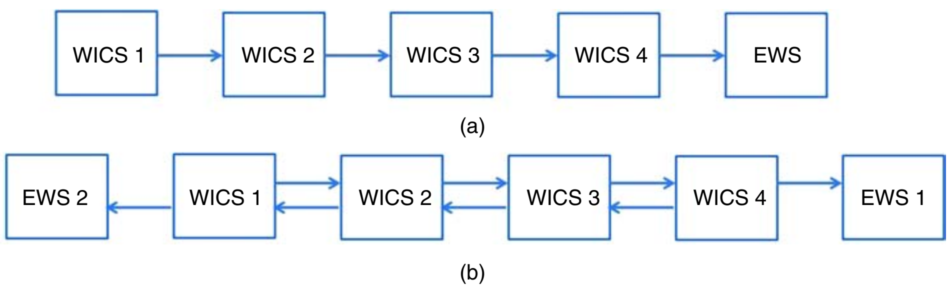

Fig. 5—Unidirectional (a) and bidirectional (b) implementations of wireless linear networks with external wireless station(s).

Once all WICSs are properly installed and powered, the one or two EWSs can monitor the activity of the pipeline on the basis of the quality of the WICN along the annular airgap. Each EWS is connected to its central unit and, in the case of multiple central units, at least one of them runs dedicated software to process the received data.

Fig. 6 illustrates the design of a pipeline prototype containing four segmentation rings properly sealed that delimit three pipe segments. Each segmentation ring hosts a WICS and the four WICSs form a unidirectional WICN.

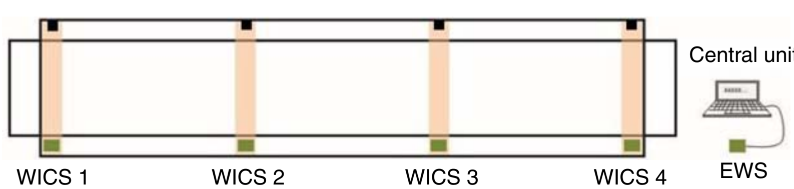

Fig. 6—Pipeline containing main pipe, protective layer, four segmentation rings defining three pipe segments, four WICSs and one EWS connected to a central unit that monitors whether the pipeline is operating properly. Lower and upper openings in each segmentation ring are denoted with squares.

If this pipeline is operating under normal conditions, the EWS receives information packets from all stations in the network. In its simplest form, each information packet contains the number of its WICS and this acknowledges that the said WICS is active. For example, WICS 1 sends information packets marked 1 to WICS 2, and WICS 2 combines these with its own ID and sends information packets 1 and 2 to WICS 3. Subsequently, WICS 3 sends information packets 1, 2, and 3 to WICS 4, which in turn sends information packets 1, 2, 3, and 4 to the EWS. In this case, the EWS is being informed that all internal WICSs are active, and that the pipeline is, therefore, operating properly.

In the case of leakage from the main pipe into the annular airgap, the annular airgap in at least one pipe segment begins to flood. For example, if the leakage starts in the pipe segment between WICS 2 and WICS 3, the wireless connection between these two WICSs is the first to be affected, and this is detected by the EWS in real time. Once this connection is affected, the number of information packets successfully sent from WICS 2 to WICS 3 starts to drop and the EWS is notified. Since WICS 2 carries information about itself and WICS 1, the number of received information packets from WICS 1 drops as well. In the extreme case, the connection between WICS 2 and WICS 3 may be lost, so that the EWS does not receive any information packet from WICS 1 and WICS 2. However, it still receives information packets from WICS 3 and WICS 4.

If no action is taken by the pipeline operator to shut down the line and fix the problem, the annular airgap of the pipe segment keeps flooding and the internal pressure builds because the small secondary opening on the top of each segmentation ring would cease serving as an air relief valve since the flooded interpipe segment would be void of air. The segmentation rings that delimit this pipe segment (and their sealants) temporarily retain the leaking fluid within the annular airgap of this pipe segment. Once the internal pressure surpasses a certain threshold value, the sealant of at least one adjacent segmentation ring breaks and the fluid starts to flood the annular airgap of a neighboring pipe segment in addition to the already leaking pressure relief valve on the top. The pressure threshold at which the sealant is designed to break is orders of magnitude lower than the maximum pressure that the protective pipe layer can withstand, so that the latter's integrity is preserved, while the leak spreads stepwise horizontally from segment to segment in the pipeline along the interpipe space until an operator reacts by reducing pipeline pressure in order to address the internal leak. Thus, if the leak persists, a second pipe segment floods and the communication between two other WICSs is also affected. For example, if the seal that breaks is that of the segmentation ring whose lower predesigned opening hosts WICS 3, then the pipe segment between WICS 3 and WICS 4 floods and communication between WICS 3 and WICS 4 is affected. In this case, the EWS receives fewer (or no) information packets from all WICSs that are positioned upstream of and including WICS 3; only information packets from WICS 4 are not affected.

In the case of a bidirectional WLN with two EWSs, each EWS monitors the WICN from one end of the line. The corresponding pipeline system is illustrated in **Fig. 7,** consisting of four WICSs and three pipe segments.

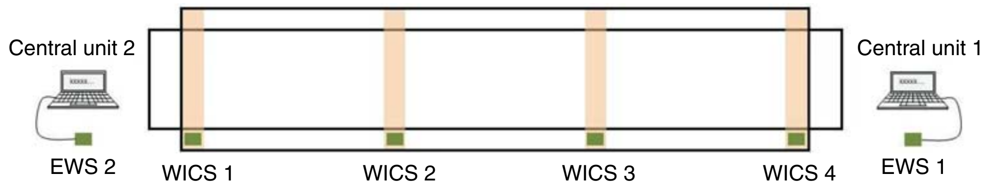

Fig. 7—Pipeline containing main pipe, protective layer, four segmentation rings defining three pipe segments, four WICSs, and two EWSs connected to two central units.

It should be mentioned that due to various circumstances (such as varying pressures and/or temperatures), the sealant of a lower predesigned opening might not break. Regardless, if the airgap in a given interpipe segment floods due to a leak from the inner pipe, leaking to the adjacent segment(s) will occur anyway from the second pressure-relieving opening located at the top of each segmentation ring. Thus, the stepwise horizontal propagation of a leak from segment to segment will be ensured even if an operator is late to react by reducing pipeline pressure in order to address the problem. This will result in the wireless information and communication system issuing multiple sequential and redundant warnings that the leak is propagating, and the direction of this propagation will be clearly known in real time. Additionally, the location of the leak with respect to the positions of the two neighboring WICS is completely irrelevant, because any leakage from the inner pipe into the interpipe space, regardless of its position, will have the very same result—gradual filling of the interpipe space between the two neighboring WICS with the leaking liquid until the wireless connection between the WICS is jeopardized to the point of interruption, the latter resulting in a permanent alarm far before the leak could reach the outside environment.

Software. After the entire WICN is configured, including all WICSs inside the pipeline and the EWS (the coordinator of the network), a dedicated software in the central unit is able to connect to its corresponding EWS. This software monitors the entire network inside the pipeline on the basis of the information packets received by the EWS (both the contents of the packets and the times of their arrivals). Then, the software processes this data and provides a meaningful display through a graphical user interface (GUI), which is designed to be interpreted by pipeline operators. In addition, the software can sound an alarm, or issue controlling commands to pressure-determining valves in order to shut off the entire pipeline, if need be.

Complete Pipeline System. The basic unit of the proposed pipeline system is a pipe apparatus containing the three major features: (a) pipe-in-pipe (see Pipe-in-Pipe Design section), (b) segmentation rings properly sealed (see Segmentation Rings section) and (c) embedded WICN connected to a monitoring software (see Embedded WICN and Internal Leakage Monitoring section). Each pipe apparatus contains interconnectable power supply lines, so that two or more pipe apparatuses of the same kind can be connected to assemble a complete pipeline system.

This basic pipe apparatus is illustrated in **Fig. 8,** and contains a main pipe (light gray), protective layer (dark gray), segmentation rings (light orange), wireless information and communication station (green), and power supply lines (red and black). One pipe apparatus defines at least one pipe segment (if one segmentation ring is used at each end of the pipe apparatus).

Fig. 8—Pipe apparatus containing main pipe, protective layer, segmentation rings, WICS, and power supply lines.

For assembly, two pipe apparatuses are positioned so that one end of each apparatus faces the other, with the main pipes aligned. The electric power supply lines of both pipe apparatuses are then connected. The protective layer is shorter than the main pipe in order to provide enough space to connect the main pipes first and the protective layers second. In the case where both pipe layers are made of metal, the adjacent main pipes are welded first, then the adjacent protective layers are encased by and welded to two external half-rings, as illustrated in **Fig. 9.**

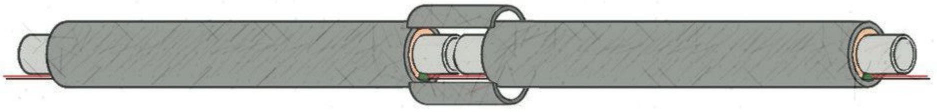

Fig. 9—Pipeline system containing two pipe apparatuses connected with the aid of two external half-rings.

It is important to highlight that the one or two central units running their respective monitoring software are part of one pipeline system, regardless of the length of the pipeline and the number of pipe apparatuses used in its assembly. This one or two central units are then responsible for monitoring the activity of the WICN within the pipeline system and for detecting any leakage into the annular airgap of one or more pipe segments.

Results

Laboratory Pipe-in-Pipe Design. In this section, a laboratory pipe-in-pipe design is described. Two steel pipes were used to build a 6 ft 8 in. pipeline system. The main pipe, rated at NPS 2.5, had the following measurements: $ID_M = 2.46$ in., $WT_M = 0.21$ in., $OD_M = 2.88$ in., and $L_M = 6$ ft 4 in. For the protective layer, the corresponding measurements were: $ID_P = 4.00$ in., $WT_P = 0.25$ in., $OD_P = 4.50$ in., and $L_P = 6$ ft 8 in. Thus, the annular airgap of the experimental pipeline had a width $W_A = 0.56$ in.

The lengths of the main pipe (L_M) and the protective layer (L_P) were not exactly the same because of a difference in the assembly of the setup of the performed tests: different connectors and adapters were designed to fit these two pipes in each test. This is explained in the following subsections, and the measurements of all connectors and adapters are given.

Segmentation Rings. In the design of the proposed leakage prevention and detection system (LPDS), two segmentation rings were used to delimit a single pipe segment of the pipeline system, and to provide support for the pipe-in-pipe design. The segmentation rings were custom designed, as illustrated in Fig. 3, and were made from Delrin® (acetal homopolymer resin; DuPont de Nemours, Inc., Wilmington, Delaware, USA), which exhibits high stiffness and strength. The measurements of the segmentation rings, as described in Fig. 3, were: $ID_R = 2.88$ in., $OD_R = 4.00$ in., $W_R = 0.56$ in., $\theta = 45°$, and $L_R = 2.00$ in.

Fig. 10 illustrates the two steel pipes used for the pipe-in-pipe design with the custom-made Delrin segmentation rings. The upper pipe is the protective layer and the lower one is the main pipe. The segmentation rings are positioned at the ends of the main pipe and the predesigned openings are aligned, before the main pipe is inserted into the protective layer.

Fig. 10—Two steel pipes used for the pipe-in-pipe design, with Delrin segmentation rings around the main pipe.

WICN. The WICN was composed of WICSs, which were implemented by XBee radio modules (Chong and Kumar 2013). The WICN used in the described laboratory setup is illustrated in **Fig. 11.** It consists of a ZigBee network, with a unidirectional WLN containing two WICS within the annular airgap of the pipeline and one EWS (the coordinator of the ZigBee network) connected to a monitoring computer.

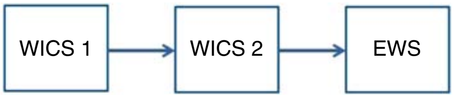

Fig. 11—Experimental linear WICN containing two WICSs and one EWS.

Each of the two WICSs located inside the pipeline was implemented using a programmable SMT XBee S2C ZigBee 2.4 GHz with a u.FL antenna (P.N. XB24CZ7UISB003). The EWS was implemented using a standard TH XBee S2C ZigBee 2.4 GHz with a connector for a u.FL antenna (P.N. XB24CZ7UIT-004), and one u.FL antenna was associated to each XBee.

Custom-designed cases were used for protection of the XBees when they were placed inside the pipeline. Two wires were soldered to the power supply pins of the XBee and inserted into two small, predrilled holes in the cover of the case. Once each SMT XBee unit was positioned inside the custom-designed case, the u.FL antenna was connected to the XBee and adjusted inside the case. Then, the entire case was sealed with epoxy and duct tape to make it watertight. The two wires corresponding to the positive supply voltage (Vcc) and ground (Gnd) were then soldered to the appropriate power supply lines within the annular airgap of the pipeline and covered with electrical tape.

The WICN illustrated in Fig. 11 was physically implemented with two programmable XBees that served as WICS 1 and WICS 2; both were configured and programed accordingly. The XBee that served as the EWS (standard XBee) was configured, but not programmed.

The described WICN setup relies on the transmission of information packets from WICS 1 to WICS 2 and from WICS 2 to EWS, as illustrated in **Fig. 12.** WICS 1 periodically sends 1 information packets to WICS 2 (one information packet every T seconds). WICS 2 receives these information packets, combines them with its own 2 information packets, and sends both types to the EWS (two information packets every T seconds). The EWS is connected to a computer running dedicated software that not only processes the data received, but also reports on the performance of the pipeline on the basis of the number and frequency of information packets received from each of the two WICSs.

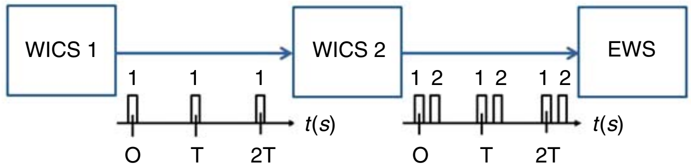

Fig. 12—Experimental WICN containing two WICSs and one EWS, with timing diagram of the information packets being sent.

Pipeline Monitoring System. Custom software was written specifically for the purpose of monitoring the activity of an experimental pipeline system. This software, named Pipeline Monitoring System (PMS), was tested on a computer running Microsoft® Windows® 7 operating system (OS), while this computer was connected to the EWS through an XBee Explorer unit and a USB cable. In order to provide a GUI, Dev C++ IDE was used to create a Windows 32-bit GUI application. The standard windows library (windows.h) was used so that the functions of the GUI could be called from within the main source code, which was written in C++.

The PMS was written to monitor the specific WICN implemented in the experimental pipeline system, so it was essential to make the software compatible with this WICN. Therefore, the number of internal WICSs was set to two, the baud rate was set to 115,200 bps and the timing diagram of information packets received at the EWS (see Fig. 12) was considered when defining the timing features in the software. Because the WICN contained two internal WICSs and one EWS, the software displayed two bar charts corresponding to the percentages of information packets received from each of the internal WICSs, as illustrated in **Fig. 13.**

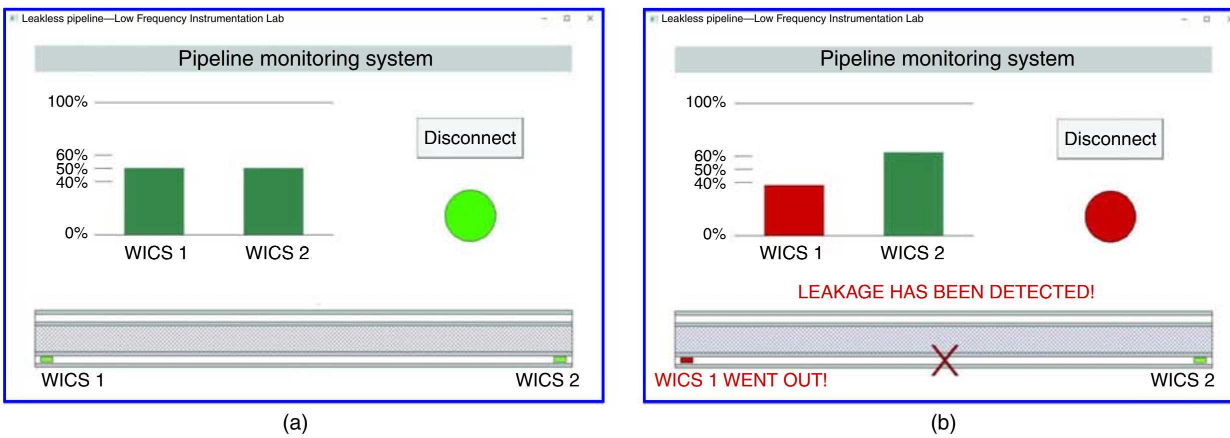

Fig. 13—Design of PMS, illustrated for the cases (a) where the pipeline is operating under normal conditions and (b) where an alarm is activated to indicate that leakage was detected.

While Fig. 13(a) illustrates the design of the main screen of this software when the pipeline is operating under normal conditions, Fig. 13(b) illustrates the case where an alarm has been activated to indicate the detection of leakage. Because only one EWS was used in this design, this EWS was responsible for receiving all information packets sent from both WICSs (information packets with 2 sent directly from WICS 2 and information packets with 1 sent from WICS 1 and forwarded by WICS 2).

Dynamic Water Testing. To demonstrate the feasibility of an actual pipeline design with an integrated leakage protection and detection system (LPDS), a comprehensive laboratory test was performed to study how the WICN behaves when water is detected in the annular airgap between the two pipe layers of the pipeline. The test consisted of a complete setup that provided dynamic water flow within the main pipe of the pipeline, which was part of a closed loop through which water could flow. Leakage was simulated so that the software could detect the leakage in real time. The minimal detectable leak amount depends on the width of the interpipe space and, more specifically, on its volume embedded between two adjacent WICS.

The purpose of the dynamic water test was to imitate the fluid transmission of a real pipeline and simulate a leak. To achieve this, the threaded caps of the main pipe were replaced with thread-to-hose adapters in order to circulate water via a hose and through the main pipe, with the aid of a water pump and a pail filled with water. Custom-made thread-to-hose adapters were built to match the existing threaded main pipe and to connect it to 1 in. plastic hoses. **Fig. 14** depicts the technical drawing of the used thread-to-hose adapter.

Initially, two segmentation rings were placed at the ends of the main pipe and their predesigned openings were aligned facing down. Both thread-to-hose adapters were connected to the ends of the main pipe, without moving the segmentation rings. Then, the power supply lines were placed on the exterior wall of the main pipe, passing through the predesigned openings of the segmentation rings. The power supply lines were made longer than the pipeline system to allow soldering to the wires connected to the XBees and the breadboard. Each power supply line had an extra 4 in. length on each end for soldering and another 2 ft at the left end for connection.

Five small holes (around 0.1 in. each) were drilled in the duct tape that protected the small cavity in the wall of the main pipe in order to simulate a leak from the main pipe into the annular airgap. Glue tape was used to firmly attach the power supply lines to the outer wall of the thread-to-hose connector of the main pipe, in order to prevent damage to the power supply lines. Once the main pipe was set, it was inserted into the protective layer, keeping the predesigned openings of the segmentation rings facing down. Then, two hoses were connected to the main pipe, one at each end, firmly positioned with the aid of a clamp. A 3 ft hose was used at the right end, and an 8 ft hose was used at the left end.

The right end of the main pipe was positioned so that the segmentation ring was aligned with the right end of the protective layer, whereas the left ends of the pipe layers had a longitudinal gap between them. This alignment between the right ends of both pipe layers is illustrated in **Fig. 15,** through a coaxial view from that end.

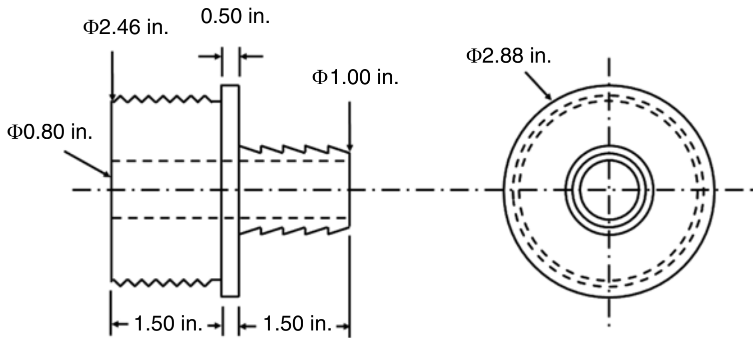

Fig. 14—Technical drawing of a thread-to-hose adapter designed to connect the main pipe to a 1 in. plastic hose.

Fig. 15—Coaxial view of the right end of the experimental pipeline used for the dynamic water test setup, illustrating the alignment between the end of the protective layer and the segmentation ring positioned at the right end of the main pipe.

Subsequently, both XBees were placed in custom-designed cases, which were sealed with epoxy. After the epoxy was dry, duct tape was applied to the entire case. Then, the XBees at both ends of the pipeline system were connected to the power supply lines, by properly soldering the wires that corresponded to Vcc (red) and Gnd (black), similar to the diagram described in **Fig. 16.** After the connection was made and isolated with electrical tape, the XBees were placed inside the predesigned openings of their segmentation rings. Power supply was provided by connecting the battery case with two AA batteries to a breadboard and properly connecting the power supply lines to the same breadboard. The ON/OFF button of the battery case was in the ON position.

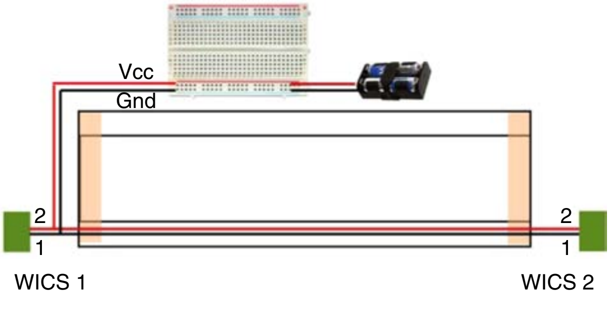

Fig. 16—Diagram of connections of the power supply lines.

The EWS was connected to the computer and the Pipeline Monitoring System was initialized. Once both bar charts corresponding to WICS 1 and WICS 2 stabilized at the 50% level, it was verified that all the XBees and the PMS were working properly. Then, the PMS was closed, the EWS was disconnected from the computer and the ON/OFF button of the battery case was set to the OFF position. Afterward, silicon paste was applied to both ends of the pipeline to entirely seal the predesigned openings of the segmentation rings. Silicon paste was also applied around the inner and outer diameters of the segmentation rings, where they touched the main pipe and the protective layer, respectively. However, a gap corresponding to roughly 90° was left on top of the segmentation ring, allowing air to pass through while the annular airgap was filled with water. A waiting period of 4 hours was required for the silicon paste to dry. Once it was verified that the paste was dry, the EWS was reconnected to the computer through its XBee Explorer, the ON/OFF button of the battery case was turned to the ON position, and the PMS software was initialized on the computer.

The unconnected end of the 3 ft hose attached to the right end of the experimental pipeline system was placed inside the pail filled with water. Another piece of hose, measuring 4 ft, was submerged in the water of the pail and the other end of the same hose was connected to the lower opening of the water pump with the aid of a clamp. Last, the unconnected end of the 8 ft hose was attached to the left end of the experimental pipeline system and connected to the upper opening of the water pump, with the aid of a clamp. The final configuration for the dynamic water test setup is shown in **Fig. 17.**

Fig. 17—Experimental setup for the dynamic water test, with pipeline system connected to a water pump and a water pail in a loop.

To verify the performance of this experiment, the water pump was turned on and immediately started draining water from the pail through its lower opening and pumping the water out through its upper opening. Due to the water pump high internal pressure, the main pipe was filled with water in less than 12 seconds (flowrate $7.192\,m^3/h$) and water pumped into the left end of the pipeline system traveled through the main pipe from left to right. Water coming out from the right end of the pipeline system was collected in the water pail, thus completing the loop.

The PMS software was initialized on the computer to which the EWS was connected and started monitoring the activity of the WICN, detecting the timing and effects of this particular leak. Once both bar charts corresponding to the percentages of information packets being received from WICS 1 and from WICS 2 were stable at 50%, it was concluded that the WICN of the experimental pipeline system was operating properly. The water pump was turned on, defining $t = 0s$ of the dynamic water test, which lasted 45 seconds.

After a few seconds, the main pipe was filled with water and, under the pressure of the water being pumped into the main pipe, small predrilled holes in the main pipe let water flow through them, simulating the leak into the annular airgap. **Fig. 18** illustrates the simulation of leakage from the main pipe into the airgap of the experimental pipeline system, and highlights the direction of water flow throughout the entire loop, in accord with the physical design of the dynamic water test presented in Fig. 17.

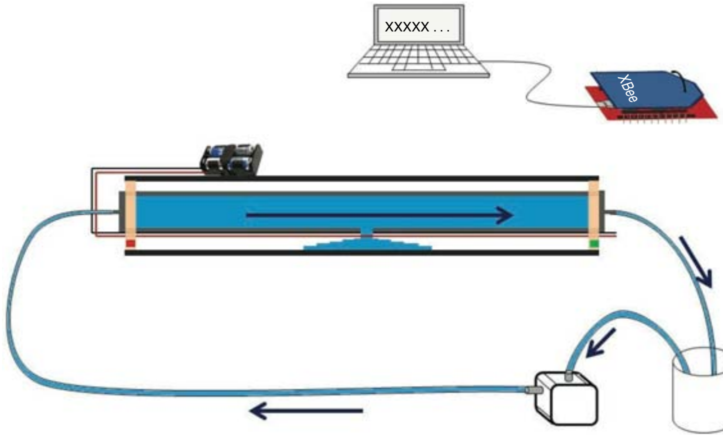

Fig. 18—Water flow and simulation of leakage from the main pipe into the annular airgap of the experimental pipeline system.

Because water was leaking into the annular airgap, the WICN was affected. Thus, at $t = 12s$, the bar chart corresponding to WICS 1 started to drop, while the chart corresponding to WICS 2 started to rise. This indicated that fewer information packets with 1 were being received at the EWS.

Regarding the occurrences of 1 and 2 information packets received at the EWS, the number of information packets received from each WICS were displayed as percentage bar charts on the screen of the PMS, instead of as strings of 1s and 2s in the serial terminal of XCTU. A decreasing number of received 1s was translated into a dropping bar chart corresponding to WICS 1. The percentage of received information packets with 2 increased and the chart corresponding to WICS 2 rose.

Because the water pump was still on, the leakage persisted and the volume of water within the annular airgap of the experimental pipeline system increased. At $t = 15s$, the bar chart of WICS 1 dropped below the 40% threshold level and turned red, as shown in **Fig. 19.** The alarm was turned on by changing the color of the alarm indicator to red and activating a buzzer. The rectangular box that represented WICS 1 in the drawing of the pipeline system also turned red, and a red cross was added to identify the faulty pipe segment in which leakage was being detected. Finally, a message—"LEAKAGE HAS BEEN DETECTED!"—was displayed in red letters on the screen to notify users of the incident and the "WICS1" label was replaced with the message "'WICS1 WENT OUT!," also in red.

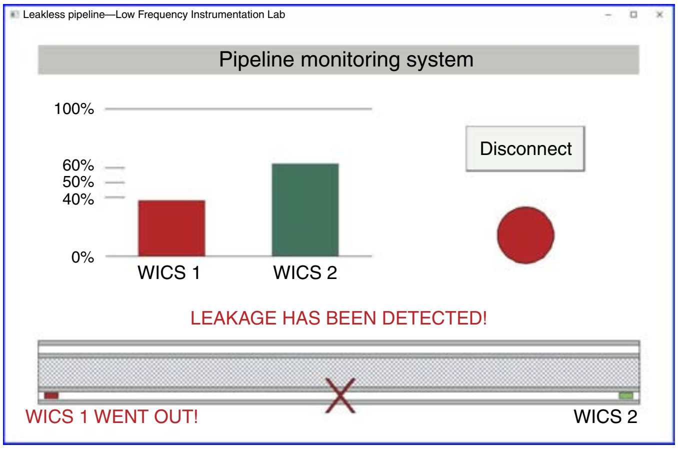

Fig. 19—The pipeline monitoring system showing the bar chart corresponding to WICS 1 dropping below the 40% level, indicating that leakage had happened.

After the alarm was activated, the bar chart corresponding to WICS 1 dropped even further for a few seconds, whereas the bar chart corresponding to WICS 2 rose. At $t = 19s$, the seals of both segmentation rings broke and water spilled from the two ends of the experimental pipeline system. Though the water spilled out of the experimental pipeline system, this test simulated only one pipe segment; in a real pipeline system, the water (or other fluid) would spread to an adjacent pipe segment without reaching the external environment. Furthermore, this spread would cause the adjacent pipe segment to flood, which would also be detected in real time by the PMS.

The dropping bar chart corresponding to WICS 1 reached a minimum at the 25% level at $t = 23s$. The spilling from both ends of the experimental pipeline system persisted and the alarm did not turn off. At $t = 25s$, the water pump was turned off, relieving the pressure inside the main pipe that was causing the leakage. The spill from the ends of the experimental pipeline system was contained and the stability of the water within the annular airgap was restored. Both bar charts then gradually returned to the normal operating region between the lower and the upper threshold levels (40–60%), and at $t = 30s$, the bar chart corresponding to WICS 1 turned green again. After the predefined alarm interval of 3 seconds, at $t = 33s$, the alarm was deactivated as the bar charts had not crossed the threshold levels again.

Despite the fact that both bar charts remained within the region between the lower and the upper threshold levels, the four indications that leakage had happened remained to indicate that leakage had been detected previously in this particular pipe segment. These indications were: (1) the red rectangle representing WICS 1, (2) the red cross in the middle of the faulty pipe segment, (3) the red message at the top of the image of the pipeline system, and (4) the red message under the image of WICS 1.

After this, both bar charts continued to be updated on the basis of incoming data received by the EWS and it was noted that the bar chart corresponding to WICS 1 varied between the 40% and 45% levels, while the chart corresponding to WICS 2 varied between the 55% and 60% levels. This behavior illustrated that though the internal leakage was contained after the water pump was turned off, water was still present in the annular airgap of the experimental pipeline system and the WICN was still affected as a result. At $t = 45s$, when the test was finished, the bar charts corresponding to WICS 1 and WICS 2 were at 40% and 60%, respectively. Because they reached these levels, but did not cross them, the alarm was not activated again.

Conclusions

Although tremendous efforts have been made in pipeline leakage detection (*RP DNV-RP-F302* 2010), pipeline leaks have resulted in continuous societal resentment toward building new pipelines. The arguments pro- and con- new pipeline construction have even reached the courts, resulting in substantial delays in oil and gas distribution across the globe. This work addresses the issue of leaking pipelines with a proposal for a leakless pipeline design and a concept of a pipeline system with leakage prevention and detection features. A combination of a pipe-in-pipe design, a WICN, and a segmented pipeline was essential for the definition of such a technology. Laboratory tests were performed in order to demonstrate the feasibility of this technology through a pipeline prototype. Custom software was written to implement a pipeline monitoring system and was coupled to the pipeline prototype to monitor its activity. The proposed technology is capable of leak prevention, which protects the environment from accidents and benefits pipeline companies.

Acknowledgments

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